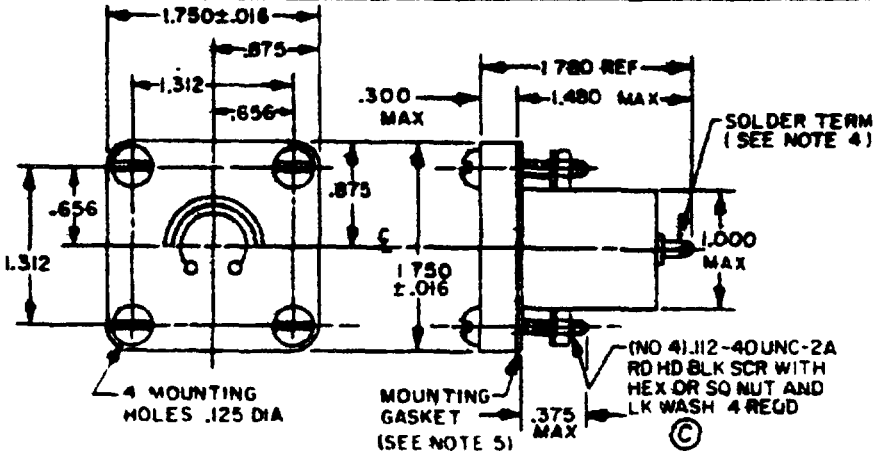
MIL-M-10304/20C
17 December 1975
SUPERSEDING
MIL-M-10304/20B
27 April 1971

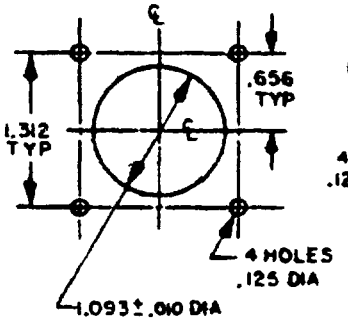
MILITARY SPECIFICATION SHEET

METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED. VOLTMETER AND AMMETER, DC (SQUARE FLANGE, 1-INCH) STYLE 03

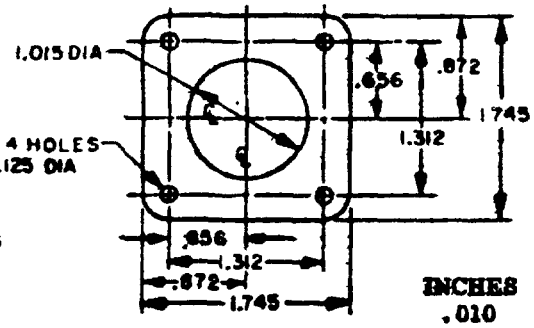
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the meters described herein shall consist of this document and the latest issue of Specification MIL-M-10304.





PANEL DRILLING PLAN



ITEM IS SYMMETRICAL ABOUT BOTH CENTER LINES

GASKET

INCHES	MM	INCHES	MM
.010	.25	1.000	25.40
.016	.41	1.015	25.78
.112	2.84	1.093	27.76
.125	3.18	1.312	33.32
.300	7.62	1.480	37.59
.375	9.53	1.745	44.32
.656	16.66	1.750	44.45
.872	22.15	1.780	45.21
.875	22.23		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm).
4. Terminal location optional.
5. Supplied with .035 (.89 mm) black rubber mounting gasket grade no. SC415 A_1 B_1 F_7 per MIL-STD-417.

© denotes changes

FSC 6625

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REQUIREMENTS:

Dimensions and configuration: See figure.

Scale
Length 1 5 inches minimum.
Pointer deflection 180° nominal.

Position influence: ±2 percent maximum change for 60° rotation from normal operating position.

Accuracy: ±2 percent.

© Overshoot· 57 percent maximum

Response time·
2 5 seconds maximum on milliammeters and voltmeters.
3 seconds maximum on microammeters.

Power consumption·
Microammeters: 750 mV maximum
Millimmeters: 150 mV maximum
Voltmeters· 1 mA ± 5 percent

High temperature cycling.
±4 percent maximum change (ammeters).
±3 percent maximum change (voltmeters).
±2 percent maximum permanent change.

Temperature influence:
± 75 percent maximum change (voltmeters).
±1 percent maximum change (ammeters).

Exposure to extreme temperature:
±3 percent maximum permanent change (ammeters and voltmeters).

Overload capacity:
Momentary overload: ±2 percent maximum permanent change.
Sustained overload:
±1.5 percent maximum temporary zero shift.
±0.5 percent maximum permanent zero shift.
±2 percent maximum permanent change.

Dielectric withstanding voltage: 1,300 volts rms.

Vibration, random drop, and shock. Repeatability is not applicable for those microammeters below 100-microampere ranges.

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TABLE. Full-scale ranges.

Voltmeters		Ammeters			
Volts		Microamperes		Milliamperes	
Zero left	Zero center	Zero left	Zero center	Zero left	Zero center
0.1	10-0-10	30	30-0-30	1	5-0-5
2	30-0-30	50	50-0-50	5	30-0-30
10	50-0-50	100	100-0-100	10	50-0-50
20		200	300-0-300	30	100-0-100
50		300	500-0-500	50	200-0-200
100		500		100	
150				200	
200				300	
				500	

Custodians:
Army - EL
Navy - EC
Air Force - 82

Review activities:
Army - EL, AV, AT, ME, WC
Navy - EC, MC
Air Force - 11
DSA - ES

User activities:
Army - MI
Navy - CG
Air Force - None

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 6625-0574-15)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-10304/20C METERS,ELECTRICAL INDICATING,PANEL TYPE, RUGGEDIZED: VOLTMETER AND AMMETER, DC(SQUARE FLANGE,1-INCH)STYLE 03

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 62 WHICH MAY BE USED

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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